Uncertainty-Aware Human Querying for Knowledge-Based Robot Planning

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Abstract

Robots operating in real-world environments must make decisions under uncertainty caused by perception errors, action execution failures, and dynamic environmental changes. Many human-in-the-loop robotic systems address this problem by relying on continuous supervision, failure-triggered intervention, or fixed query policies, which can increase operator workload or request assistance only after errors have occurred. This paper presents a knowledge-based robot planning framework that determines both *when* to ask a human and *what* information to ask for during task execution. The proposed framework combines a STRIPS-style symbolic knowledge base with a belief distribution over possible world states and uses entropy-based uncertainty estimation to trigger human queries. When uncertainty remains high after an observation, the system selects a symbolic hypothesis whose answer is expected to maximally reduce belief uncertainty, obtains lightweight human feedback, and updates the knowledge base before replanning. We integrate the framework into a complete robotic architecture connecting sensing, planning, feedback management, user interaction, and knowledge management. The system is evaluated in tomato harvesting and waste sorting domains using simulation, real-robot experiments, and user studies. The results demonstrate that uncertainty-aware querying reduces unnecessary human interaction and cognitive workload while maintaining task performance.

Keywords: autonomous robots, belief-space planning, human-robot interaction, knowledge-based robotics, POMDP, proactive querying, planning under uncertainty

1 Introduction

Recent advances in robotic systems have significantly improved autonomy in perception, manipulation, and planning across domains such as agriculture and household environments. Despite these advances, robots still frequently fail in real-world environments due to perception errors [22], action execution failures [21], and dynamic environmental changes [26]. Consequently, many robotic systems still rely on human operators to continuously monitor execution and intervene when failures occur [10, 13]. However, requiring humans to continuously monitor robotic systems

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Figure 1: Overview of the proposed knowledge-based robotic system with uncertainty-aware planning. The system maintains a belief over possible worlds and updates it based on observations. If uncertainty exceeds a threshold, the system selectively queries the user to resolve ambiguity; otherwise, it continues autonomous planning and execution.

and assess the reliability of robot decisions is impractical for long-horizon human–robot collaboration because it increases cognitive workload and fatigue [4]. Instead of relying on humans to identify failures after they occur, robotic systems should be able to recognize uncertain situations on their own and proactively request assistance only when necessary [45, 29]. This shift leaves two coupled questions unresolved: (i) *when* should the robot ask a human, and (ii) *what* should it ask?

To answer these questions, the robot must estimate its uncertainty about the world and determine whether additional human input is likely to improve decision making. We therefore formulate robotic decision making as a POMDP [21] and develop a framework that jointly models symbolic knowledge [8] and uncertainty within a unified structure. We adopt a STRIPS-style symbolic representation [12], where the knowledge base is defined as a set of explicit facts, while uncertainty is represented as a belief distribution over possible world states corresponding to alternative hypotheses about the environment. As observations are collected, the system updates the belief over these states accordingly. By measuring the entropy of this belief distribution, the system estimates uncertainty and determines *when* to query a human. The system also determines *what* to ask by selecting questions that are expected to maximally reduce uncertainty among competing hypotheses. Since interaction only requires simple feedback about the environment state, the proposed framework reduces user cognitive burden without requiring users to make planning decisions.

We further design and integrate a system architecture that enables the proposed framework to operate on a real robotic platform. The architecture maintains a belief over possible world states and updates it using both sensor observations and human feedback. A centralized knowledge base connects the sensing, planning, feedback management, and user interface modules, allowing information to be shared consistently across the system. This design enables the robot to reason with the most up-to-date information and request human assistance only when uncertainty arises.

We evaluate the proposed system in two domains, tomato harvesting and waste sorting, using both simulation and real-robot experiments. Experimental results show that the proposed approach reduces the number of human interactions while maintaining or improving task success rates. Furthermore, user studies demonstrate that the system effectively reduces cognitive workload during human–robot collaboration.

The main contributions of this work are as follows:

- We propose a knowledge-based robotic framework that jointly represents symbolic world knowledge and uncertainty over possible world states, enabling decision making under partial observability while preserving interpretability.
- We present an uncertainty-aware human feedback mechanism that addresses both *when* and *what* to ask by estimating belief uncertainty and selecting informative queries that distinguish competing hypotheses.
- We design and integrate a complete robotic system that combines sensing, planning, user feedback, and knowledge management through a centralized knowledge base.
- We validate the proposed framework in both tomato harvesting and waste sorting domains through simulation, real-robot experiments, and user studies, demonstrating reduced human intervention and cognitive workload while maintaining task performance.

2 Related Work

In this section, we review prior work on knowledge-based robot planning, belief-space planning under uncertainty, human involvement in robot autonomy, and uncertainty-aware query systems.

2.1 Knowledge-Based Robot Planning

Knowledge-based robotic systems use structured representations to describe objects, properties, relations, actions, and task constraints. Logic-based knowledge systems and robot knowledge bases have been used to store observed facts, infer missing information, and support semantic reasoning in robot tasks [53, 33]. For example, robots can infer likely object locations from qualitative spatial relations, container-location relations, or place-object co-occurrence knowledge [25, 15]. Commonsense and relational reasoning further connect symbolic knowledge with perceptual evidence in partially observable environments [37, 58, 7]. Ontologies and type hierarchies support consistent integration of new observations and generalization across related objects, actions, and task contexts [36, 40, 35].

Classical symbolic planning provides a computational layer on top of such knowledge representations. STRIPS and PDDL represent states as sets of symbolic facts and actions as preconditions and effects [12, 1]. Related planning representations such as FDR and domain-independent planners expose compact state variables and enable efficient search over symbolic transition systems [17, 18, 16, 5]. This structure is useful in robotics because high-level actions can be checked against symbolic preconditions and then grounded into executable robot skills. However, classical symbolic planning typically assumes that the current state is known, whereas real robot knowledge is often produced by noisy perception and partial observations.

Recent work has expanded robot knowledge beyond hand-coded symbolic rules using large language models and vision-language models. LLM-based planning systems use procedural and commonsense knowledge to propose task decompositions or action sequences, which are then grounded with robot affordances, programmatic interfaces, or execution feedback [2, 20, 50, 30, 9].

Other systems construct richer environment knowledge from scene graphs and open-vocabulary spatial representations, allowing robots to reason about objects, places, and relations beyond a closed symbolic vocabulary [14, 28]. Retrieval-augmented methods use demonstrations, procedural knowledge, or task-relevant constraints to guide long-horizon robot planning [51, 60, 57, 54, 24]. Related approaches combine LLM-generated plans with symbolic constraints, spatial reasoning, or verification mechanisms to improve executability and consistency [19, 6, 32, 42, 43].

These systems show that modern knowledge sources can greatly broaden what robots know and how they generate plans. Nevertheless, both traditional knowledge bases and foundation-model-based systems often commit inferred knowledge as a proposal, prior, or verified fact rather than maintaining an explicit execution-time belief over alternative symbolic worlds. Their outputs may remain uncertain because of perception errors, incomplete grounding, missing observations, or hallucinated semantic assumptions. Our work keeps the interpretability and action structure of symbolic planning while representing uncertain local facts as a belief over possible symbolic states. This lets the planner reason about what it does not know before committing perceptual outputs or inferred facts to the knowledge base.

2.2 Belief-Space Planning and Symbolic Uncertainty

POMDPs provide a standard framework for decision making under partial observability [21]. Instead of assuming access to the true state, a POMDP maintains a belief distribution over possible states and selects actions to maximize expected return. Online methods such as POMCP scale this idea by using Monte Carlo tree search and particle beliefs over action-observation histories [49]. Related online POMDP methods have been applied to robot navigation, object search, manipulation, exploration, and interaction under uncertainty [52, 27, 56, 41].

These methods show that particle beliefs are effective when the robot must preserve multiple hypotheses about the world. However, many POMDP formulations operate over task-specific state variables rather than explicit symbolic knowledge bases. Conversely, symbolic planners expose action preconditions and logical structure, but often lack a principled belief representation. Prior work on symbolic planning under uncertainty addresses this gap by combining logical structure with probabilistic inference or POMDP-style planning [46, 59, 48]. Such methods demonstrate that symbolic structure can help organize uncertainty, but uncertainty is usually resolved through autonomous sensing, transition models, or planning abstraction.

Our setting adds a different source of information: human semantic feedback. Some ambiguities cannot be reliably resolved by additional robot sensing alone, such as visually ambiguous object states or uncertain task-relevant categories. We therefore maintain a symbolic particle belief over reachable frontier states and allow the robot to query a human when the belief is insufficiently confident. The answer directly filters the planner's belief and updates the knowledge base.

2.3 Human Involvement in Robot Autonomy

Human involvement has long been used to improve robot reliability under uncertainty. In teleoperation and supervisory control, humans monitor the robot and intervene when needed [13, 10]. Human-on-the-loop systems preserve robot autonomy during nominal operation while relying on an operator to detect and correct failures [47, 39, 23, 34]. These systems can improve safety and robustness, but they require sustained human attention. The operator must notice that a problem exists, interpret the situation, and decide how to intervene, which can increase workload and reduce scalability in long-horizon or repetitive tasks [11, 55].

Another approach is failure-triggered assistance, where the robot asks for help only after it fails, reaches a dead end, or detects an invalid state. This reduces continuous monitoring, but it treats human input as recovery rather than prevention. In tasks with irreversible or costly actions, asking after failure may be too late. A third approach is frequent or fixed querying, where the robot asks whenever a predefined uncertainty condition is met. This can improve task success, but it often produces unnecessary interactions because it does not distinguish uncertainty that matters for the current plan from uncertainty that is irrelevant.

Our work targets a proactive middle ground. The human is not asked to continuously supervise the robot, and the robot does not wait until failure. Instead, the robot estimates whether its current symbolic belief is confident enough to support planning. If not, it asks a targeted question about the symbolic fact expected to reduce the most uncertainty.

2.4 Uncertainty-Aware Querying

Recent work has studied robots and embodied agents that ask for help based on uncertainty. Active learning and active preference learning query humans to improve models or infer preferences [45]. Proactive robot behavior has also been shown to improve collaboration when the robot can anticipate when assistance is useful [3]. More recent uncertainty-aware systems use model confidence, language-model uncertainty, or conformal prediction to decide when to request human input [44, 31, 38]. These methods are important because they move beyond continuous supervision and show that selective querying can reduce unnecessary intervention.

However, many query systems focus on action-level uncertainty or model-output confidence. In such systems, the human may be asked which action the robot should take, whether a proposed action is acceptable, or whether a model prediction should be trusted. This differs from asking about the latent symbolic state that causes planning uncertainty. If the robot asks for an action, the human effectively participates in planning. If the robot asks for a state fact, the human provides information while the robot remains responsible for planning.

Our framework follows the second approach. The robot asks predicate-level questions about task-relevant symbolic facts, such as object properties, perceived categories, or local symbolic conditions. Human responses eliminate inconsistent hypotheses from the belief and update the knowledge base, after which planning proceeds autonomously. By integrating selective querying with symbolic belief maintenance and online planning, our approach treats human feedback as an information source for reducing planning uncertainty rather than as a mechanism for action selection.

3 Preliminaries

3.1 Symbolic Planning

Let $\mathcal{F}$ be a finite set of symbolic atoms. A symbolic state is a subset $s \subseteq \mathcal{F}$. An action $a \in \mathcal{A}$ is defined by preconditions, add effects, and delete effects:

$$a = \langle \mathrm{pre}(a), \mathrm{add}(a), \mathrm{del}(a) \rangle.$$

The action is applicable in state s if $\mathrm{pre}(a) \subseteq s$. Applying a produces

$$s' = (s \setminus \mathrm{del}(a)) \cup \mathrm{add}(a).$$

In robot domains, symbolic actions are grounded to executable skills such as detecting an object, navigating to a location, picking an object, placing an object, scanning a tomato, or discarding an item.

3.2 POMDPs and POMCP

A POMDP is a tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, T, R, \Omega, O, \gamma \rangle$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $T(s'|s,a)$ the transition model, $R(s,a)$ the reward function, Ω the observation space, $O(o|s',a)$ the observation model, and γ the discount factor. Since the state is not directly observed, the robot maintains a belief $b(s)$ over states. After action a and observation o, the Bayesian belief update is

$$b'(s') = \eta O(o|s',a) \sum_{s \in \mathcal{S}} T(s'|s,a) b(s), \tag{1}$$

where η is a normalizer.

POMCP approximates online POMDP planning through Monte Carlo tree search over histories [49]. Each history node stores sampled particles and action values. During simulation, actions are selected using an upper-confidence rule,

$$a^* = \arg\max_a \left[V(ha) + c\sqrt{\frac{\log N(h)}{N(ha)}} \right], \tag{2}$$

where $V(ha)$ is the estimated value of taking action a after history h, $N(h)$ is the visit count of the history, and $N(ha)$ is the visit count of the action branch.

3.3 Human Feedback as Information

We treat a human response as an observation over a symbolic hypothesis. A hypothesis is a literal $q \in \mathcal{F}$ whose truth value is uncertain in the current belief. A query asks the human to verify q, producing an answer $y \in \{\texttt{true}, \texttt{false}\}$. The answer filters the belief by retaining states consistent with y. Thus, a query affects future planning by changing the posterior over symbolic worlds.

4 Method

4.1 Problem Formulation

The robot is given a symbolic domain, a set of executable skills, an initial knowledge state, and a task goal. Some facts are known with certainty, while local task facts must be inferred from observations. The robot selects task actions $a \in \mathcal{A}$ using POMCP. After each action, the observation may support several possible successor states rather than a single deterministic outcome. The system then computes belief uncertainty and requests human feedback only if the updated belief is not confident enough to commit a symbolic state to the knowledge base. The objective is to maximize task reward while reducing unnecessary feedback requests:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{T} \gamma^t \left(R(s_t, a_t) - \lambda \mathbb{I}[\mathrm{query}_t] \right) \right], \tag{3}$$

where query_t indicates that feedback is requested after the belief update at time t, and λ is the cost of human interaction.

4.2 Knowledge Base

The knowledge base is partitioned into domain information, static information, and local information:

$$\mathcal{KB} = \mathcal{I}^D \sqcup \mathcal{I}^S \sqcup \mathcal{I}^L. \tag{4}$$

Domain information encodes predicates, action schemas, task rules, and robot skills. Static information contains facts and numeric fluents that remain fixed during an episode, such as object identifiers, bin locations, or stem locations. Local information contains facts inferred during execution, such as object category, tomato state, object pose, and visibility. Uncertainty is concentrated in $\mathcal{I}^L$ because these facts depend on perception and partial observations.

4.3 Symbolic Frontier Belief

Let s^K denote the current committed knowledge state. Instead of constructing a global belief over all symbolic states, we construct a local belief over states reachable from s^K under the selected task action. For action a, the symbolic frontier is

$$\mathrm{Frontier}(s^K, a) = \{s_1^\star, \ldots, s_j^\star\}, \tag{5}$$

where each $s_k^\star$ is a reachable successor hypothesis consistent with the action model and the current knowledge base. The belief over the frontier is represented by weighted particles:

$$\mathcal{B} = \{(s_k^\star, w_k)\}_{k=1}^{j}, \qquad \sum_{k=1}^{j} w_k = 1. \tag{6}$$

After executing a and receiving observation o, each particle weight is updated by the observation model:

$$w_k' = w_k O(o | s_k^\star, a), \qquad \tilde{w}_k = \frac{w_k'}{\sum_{\ell=1}^{j} w_\ell'}. \tag{7}$$

This local frontier construction has two roles. First, it avoids enumerating irrelevant symbolic worlds. Second, it ties uncertainty to the action currently affecting task progress, which makes later human queries task-relevant.

4.4 Confidence and Query Trigger

We quantify uncertainty using the entropy of the normalized frontier belief:

$$H(\mathcal{B}) = -\sum_{k=1}^{j} \tilde{w}_k \log_2 \tilde{w}_k. \tag{8}$$

The confidence score is

$$\mathrm{Conf}(\mathcal{B}) = \begin{cases} 1, & j = 1, \\ 1 - \frac{H(\mathcal{B})}{\log_2 j}, & j > 1. \end{cases} \tag{9}$$

If $\mathrm{Conf}(\mathcal{B}) \geq \tau$, the robot commits the maximum-weight state to the knowledge base:

$$s^K \leftarrow \arg\max_{s_k^\star \in \mathcal{B}} \tilde{w}_k. \tag{10}$$

If $\mathrm{Conf}(\mathcal{B}) < \tau$, the robot requests human feedback. The threshold τ is therefore an autonomy-interaction parameter: lower values allow the robot to act with less certainty, while higher values increase interaction to obtain more reliable knowledge.

4.5 Query Selection

The system constructs candidate hypotheses from literals whose truth values differ across the frontier states. Let $\mathcal{C}$ be this set. For each candidate $c \in \mathcal{C}$, the belief is partitioned into states where c is true and false:

$$\mathcal{B}_c^+ = \{(s_k^\star, \tilde{w}_k) \mid c \in s_k^\star\}, \quad \mathcal{B}_c^- = \{(s_k^\star, \tilde{w}_k) \mid c \notin s_k^\star\}.$$

The selected query minimizes expected posterior entropy:

$$c^* = \arg\min_{c \in \mathcal{C}} \left[P(c) H(\mathcal{B}_c^+) + P(\neg c) H(\mathcal{B}_c^-) \right], \tag{11}$$

where $P(c) = \sum_{s_k^\star : c \in s_k^\star} \tilde{w}_k$ and $P(\neg c) = 1 - P(c)$. After the human answers, the belief is filtered:

$$\mathcal{B} \leftarrow \begin{cases} \mathcal{B}_{c^*}^+, & y = \texttt{true}, \\ \mathcal{B}_{c^*}^-, & y = \texttt{false}. \end{cases} \tag{12}$$

The remaining weights are renormalized. Querying repeats until the confidence exceeds τ or no informative hypothesis remains.

4.6 Planning and Execution Pipeline

At each step, the planning manager runs POMCP using the current belief and the symbolic action constraints. The selected task action is executed by a corresponding robot skill. Perception produces observations used to update the frontier belief. If the updated confidence is low, the feedback manager converts the selected symbolic hypothesis into a human-readable query, highlights the relevant object in the interface, receives a binary response, and returns the response to the planner. The knowledge base is updated only after the belief is sufficiently confident.

Figure Placeholder

fig:architecture

Figure 2: Detailed system architecture connecting the knowledge base, sensing and perception module, POMCP planning manager, feedback manager, user interface, and robot execution layer.

Figure Placeholder

fig:domains

Figure 3: Experimental domains used to evaluate the proposed framework, including simulated waste sorting and tomato harvesting with real-robot validation.

5 Experiments

The experiments are designed to answer four questions. First, does uncertainty-aware querying maintain task performance while reducing human interaction? Second, is the timing and content of queries important compared with random or fixed query policies? Third, how does the confidence threshold τ control the trade-off between autonomy and interaction? Fourth, does selective querying reduce human workload compared with continuous supervision or action-level intervention?

5.1 Domains

Waste sorting. The simulated waste sorting domain consists of a UR5 robot sorting objects into category-specific bins. Each object belongs to one of four categories: general waste, paper, can, or plastic. Categories are not fully known in advance and may be ambiguous due to occlusion or perception uncertainty. The robot must detect, pick, and place objects into the correct bins.

Input: initial knowledge state s_0^K, threshold τ
Output: executed action sequence
$s^K \leftarrow s_0^K$;
for $t = 0$ **to** T **do**
  $a_t \leftarrow \mathrm{POMCP}(s^K, \mathcal{B})$;
  if $a_t = \emptyset$ **then**
    break
  end
  execute a_t and receive observation o_{t+1};
  construct $\mathrm{Frontier}(s^K, a_t)$ and update $\mathcal{B}$ using Eq. 7;
  compute $\mathrm{Conf}(\mathcal{B})$ using Eq. 9;
  while $\mathrm{Conf}(\mathcal{B}) < \tau$ **do**
    select c^* using Eq. 11;
    query human for c^* and receive y;
    filter $\mathcal{B}$ using Eq. 12;
    recompute $\mathrm{Conf}(\mathcal{B})$;
  end
  commit the most likely state to s^K using Eq. 10;
end

Algorithm 1: Uncertainty-Aware Planning with Human Querying

Tomato harvesting. The tomato harvesting domain contains tomatoes attached to stems. Each tomato may be ripe, unripe, or rotten. Ripe tomatoes should be harvested, rotten tomatoes should be discarded, and unripe tomatoes should be left untouched. The real-robot platform consists of a mobile base, a UR5e arm, an RGB-D camera, and a LiDAR sensor. This domain evaluates whether the framework can connect symbolic uncertainty, real perception, human feedback, and robot execution.

5.2 Baselines and Ablations

We compare the proposed method against the following strategies:

- **No Query**: the robot never asks for human feedback.
- **Always Query**: the robot queries at every uncertain decision point, approximating full human supervision.
- **Random Query**: the robot queries randomly with the same average query budget as the proposed method.
- **Failure-Triggered Query**: the robot asks only after reaching a failure, dead end, or invalid action.
- **Ours without Frontier**: the robot uses confidence thresholding but selects queries without action-relevant frontier structure.
- **Ours**: the full uncertainty-aware frontier query framework.

Table 1: System evaluation across waste sorting and tomato harvesting domains.

Domain	Method	Success (%)	Reward	Queries	Time (s)
Waste	No Query	–	–	–	–
	Always Query	–	–	–	–
	Random Query	–	–	–	–
	Failure-Triggered	–	–	–	–
	Ours w/o Frontier	–	–	–	–
	Ours	–	–	–	–
Tomato	No Query	–	–	–	–
	Always Query	–	–	–	–
	Random Query	–	–	–	–
	Failure-Triggered	–	–	–	–
	Ours w/o Frontier	–	–	–	–
	Ours	–	–	–	–

5.3 Metrics

We report task success rate, cumulative reward, average number of queries, query ratio, task completion time, and failure type. For user studies, we additionally report subjective workload using NASA-TLX and interaction preference ratings. For real-robot evaluation, we report scenario-level success, number of queries, intervention timing, and representative failure cases.

5.4 System Evaluation

The results compare task performance and interaction cost across query strategies. Always Query is expected to achieve high success at the cost of frequent interaction, whereas No Query may fail when ambiguity remains unresolved. Random Query often spends queries on non-critical states. The proposed method aims to approach the task success of Always Query with fewer human queries by targeting uncertainty that is relevant to planning.

5.5 Threshold Analysis

The confidence threshold τ controls the autonomy-interaction trade-off. Low τ values make the robot more willing to commit uncertain beliefs, reducing queries but increasing failure risk. High τ values make the robot ask more often, improving confidence but increasing workload. We analyze success rate, reward, and query count across τ values in both domains because τ is the key parameter governing selective autonomy.

5.6 Real-Robot Tomato Evaluation

The real-robot study includes repeated trials across multiple tomato scenarios. Each scenario specifies the initial arrangement, tomato states, visual ambiguities, and expected task outcome. The

Figure Placeholder

fig:tau

Figure 4: Threshold sensitivity for τ across waste sorting and tomato harvesting domains.

Table 2: User study results comparing interaction conditions.

Condition	NASA-TLX	Responses	Response Time	Preference
Always Query	–	–	–	–
Failure-Triggered	–	–	–	–
Action-Level Query	–	–	–	–
Ours	–	–	–	–

evaluation reports whether each query prevents a wrong harvest, wrong discard, or missed target. A qualitative case study compares the same scene under No Query and Ours, showing how a human answer changes the belief and the subsequent plan.

5.7 User Study

The user study evaluates whether system-initiated state queries reduce workload compared with continuous supervision and action-level assistance. Participants observe or interact with the robot through the interface while the robot executes task scenarios containing perceptual ambiguity. We compare at least three interaction conditions: **Always Query**, where the user is repeatedly asked to verify uncertain information; **Failure-Triggered Query**, where the user is asked only after the robot reaches an invalid or failed state; and **Ours**, where the system asks targeted state-level questions when entropy-based confidence is low.

We measure subjective workload using NASA-TLX, along with the number of user responses, response time, perceived interruption, trust, and preference. The main hypothesis is that the proposed method reduces workload because the user is not required to continuously monitor the robot or make action-level planning decisions. Instead, the user answers targeted questions about symbolic state facts, while the robot remains responsible for belief update and replanning.

6 Discussion

The central claim of this paper is that asking a human should be treated as part of robot planning, not as an external recovery behavior. This matters because the usefulness of a query depends on the robot's current belief and on the task consequences of remaining uncertain. A query about a visually ambiguous tomato is valuable if the answer changes whether the robot harvests, discards, or ignores it. The same query may be unnecessary if the tomato is irrelevant to the current goal or if another action can resolve the uncertainty autonomously.

The frontier belief representation is intended to make this connection explicit. By constructing uncertainty around states reachable from the current knowledge state and selected action, the robot avoids asking about facts that do not affect near-term planning. The entropy-based confidence score determines when the robot lacks enough information to commit a state, and the expected-entropy query rule determines which symbolic fact would most reduce the ambiguity.

Several limitations should be discussed clearly. First, the framework assumes that human answers are reliable or at least more reliable than the robot's uncertain observation. Future work should model noisy human feedback. Second, the current query form is binary and predicate-level. More expressive questions may reduce the number of interactions but increase user burden. Third, the planner uses a local frontier belief rather than a full global posterior. This improves tractability but may miss dependencies outside the immediate action-relevant frontier. Finally, the real-robot evaluation must show repeated trials rather than isolated demonstrations for the method to be convincing as a robotics contribution.

7 Conclusion

We presented a knowledge-based belief-space planning framework that integrates proactive human querying into robot decision making. The robot maintains uncertainty over symbolic possible worlds, uses POMCP for online planning, and requests human feedback only when the observation-updated belief remains insufficiently confident. By selecting queries that are expected to reduce entropy over task-relevant frontier states, the system decides when and what to ask after ambiguous observations. This formulation uses human feedback as a belief-refinement mechanism rather than as action-level control, enabling robots to reduce unnecessary supervision while resolving ambiguity before it causes task failure. The user study further evaluates whether this selective state-level interaction reduces subjective workload compared with continuous supervision or action-level assistance.

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A Appendix

A.1 Experiment Settings

This appendix should list the exact planner parameters, perception parameters, domain parameters, and scenario descriptions used in the waste sorting and tomato harvesting experiments.

A.2 Additional Threshold Results

Additional per-scenario threshold plots and raw statistics for τ should be placed here after the main threshold analysis is summarized in Sec. 5.